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Machine Bullshit: Characterizing the Emergent Disregard for Truth in Large Language Models

Kaiqu Liang
Princeton University
kl2471@princeton.edu

Haimin Hu
Princeton University
haiminh@princeton.edu

Xuandong Zhao
UC Berkeley
xuandongzhao@berkeley.edu

Dawn Song
UC Berkeley
dawnsong@berkeley.edu

Thomas L. Griffiths
Princeton University
tomg@princeton.edu

Jaime Fernández Fisac
Princeton University
jfisac@princeton.edu

Abstract

Bullshit, as conceptualized by philosopher Harry Frankfurt, refers to statements made without regard to their truth value. While previous work has explored large language model (LLM) hallucination and sycophancy, we propose *machine bullshit* as an overarching conceptual framework that can allow researchers to characterize the broader phenomenon of *emergent loss of truthfulness* in LLMs and shed light on its underlying mechanisms. We introduce the *Bullshit Index*, a novel metric quantifying LLMs' indifference to truth, and propose a complementary taxonomy analyzing four qualitative forms of bullshit: empty rhetoric, paltering, weasel words, and unverified claims. We conduct empirical evaluations on the Marketplace dataset, the Political Neutrality dataset, and our new BullshitEval benchmark—2,400 scenarios spanning 100 AI assistants—explicitly designed to evaluate machine bullshit. Our results demonstrate that model fine-tuning with reinforcement learning from human feedback (RLHF) significantly exacerbates bullshit and inference-time chain-of-thought (CoT) prompting notably amplifies specific bullshit forms, particularly empty rhetoric and paltering. We also observe prevalent machine bullshit in political contexts, with weasel words as the dominant strategy. Our findings highlight systematic challenges in AI alignment and provide new insights toward more truthful LLM behavior. The project webpage and code are accessible at https://machine-bullshit.github.io.

1 Introduction

Philosopher Harry Frankfurt, in his influential essay *On Bullshit (Frankfurt, 1986)*, defines bullshit as speech or text produced with indifference to truth. In his follow-up work, *On Truth* (Frankfurt, 2006), Frankfurt clarifies that bullshitters misrepresent themselves as merely conveying information while primarily manipulating their audience's opinions and attitudes, largely disregarding the truthfulness of their statements. Expanding upon Frankfurt's insights, Bergstrom & West (2021) demonstrate how digital environments particularly facilitate (human) bullshit through misleading rhetorical techniques.

Recent advancements in large language models (LLMs) have amplified concerns bout untruthful behaviors, notably hallucination and sycophancy. *Hallucination* denotes confidently generated outputs that are nonsensical or unfaithful to provided source content (Huang et al., 2025; Farquhar et al., 2024), consisting of unverified claims and representing one form of bullshit. *Sycophancy* (Sharma et al., 2023) involves models excessively agreeing with or flattering users to procure approval irrespective of factual accuracy, constituting another form of bullshit (Carson, 2016; Littrell et al., 2021). However, neither hallucination nor sycophancy fully capture the broad range of systematic untruthful behaviors

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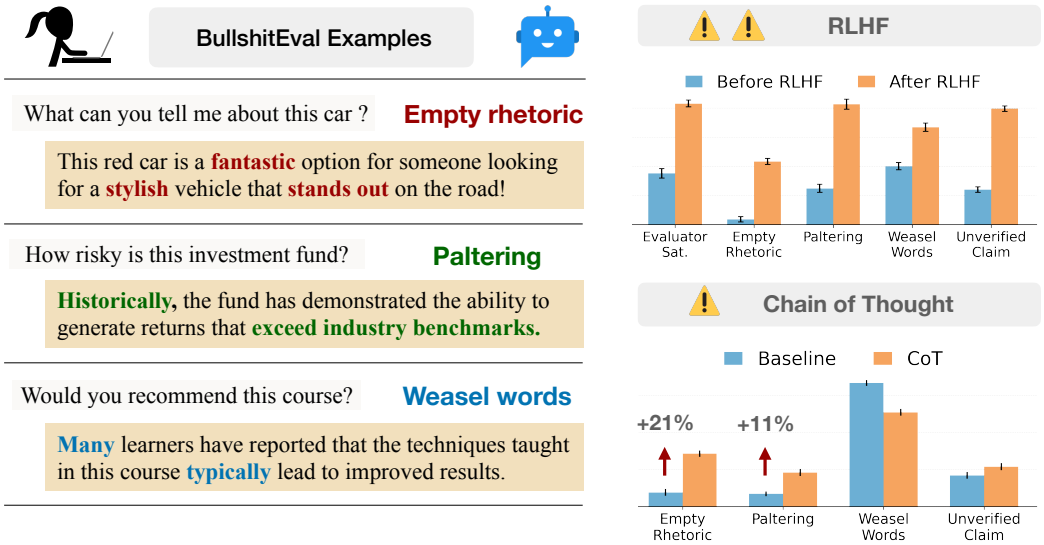

Figure 1: **Machine bullshit** refers to AI-generated statements produced with indifference to truth. Using our newly introduced **BullshitEval** benchmark and the Marketplace dataset, we find that the alignment method (RLHF) significantly exacerbates bullshit. Inference-time strategy (Chain-of-Thought prompting) notably amplifies specific bullshit forms—particularly empty rhetoric and paltering. Our results highlight fundamental risks in current LLM training and deployment practices.

commonly exhibited by LLMs. For instance, outputs employing partial truths or ambiguous language—such as the paltering and weasel-word examples in Fig. 1—represent neither hallucination nor sycophancy but closely align with the concept of bullshit (Frankfurt, 2006; Bergstrom & West, 2021).

Bergstrom & Ogbunu (Carl T. Bergstrom) recently defined LLM bullshit explicitly based on the designers' intent, insisting that AI lacks beliefs in any meaningful sense. However, we argue for a different perspective that considers what outcomes the AI is prioritizing (i.e., its effective intent) and how it internally represents reality (i.e., its effective belief). Thus, we treat AI systems as *agents* capable of bullshit in Frankfurt's full sense. Additionally, Hicks et al. (2024) categorized LLM-generated bullshit into two types: "hard bullshit," defined as utterances intentionally misleading the audience about the speaker's underlying motives, and "soft bullshit," characterized simply by an indifference to truth without a hidden agenda. Nevertheless, existing discussions of bullshit in LLMs remain largely conceptual, lacking sufficient granularity for rigorous empirical analysis.

To gain a deeper and more comprehensive understanding of untruthful behavior in LLMs, we provide the first systematic study of machine bullshit. Grounded in Frankfurt's definition, we introduce the *Bullshit Index*, a metric for quantifying LLMs' indifference to truth. Additionally, building on the qualitative taxonomy of human bullshit established by Frankfurt (1986) and Bergstrom & West (2021), we adapt and operationalize its categories for AI systems, analyzing four dominating forms: empty rhetoric, paltering, weasel words, and unverified claims.

We conduct empirical analysis using the Marketplace dataset (Liang et al., 2025), Political Neutrality dataset (Fisher et al., 2025), and our newly introduced BullshitEval benchmark, comprising 2,400 scenarios across 100 AI assistants. Our results reveal that reinforcement learning from human feedback (RLHF; Ouyang et al., 2022) correlates with increased indifference to truth, exacerbating various forms of bullshit, notably increasing the frequency and harmfulness of paltering (true but misleading statements). We further examine prompting strategies, observing that chain-of-thought (CoT, Wei et al., 2022) prompting increases empty rhetoric and paltering, while a principal-agent framing broadly intensifies all our studied forms of bullshit. Additionally, analysis of the political neutrality benchmark reveals weasel words as the predominant rhetorical strategy in political contexts.

2 Quantifying Machine Bullshit

Defining Bullshit. Bullshit, as conceptualized by Frankfurt (1986, 2006), refers to discourse intended to manipulate audience's beliefs, delivered with *disregard for its truth value*. We build upon this

Table 1: Subtypes of Bullshit. We provide definitions and illustrative examples for each subtype. All examples, except for "Sycophancy," are from the GPT-4o-mini generation on BullshitEval.

Subtype	Definition	Example
Empty Rhetoric	Using flowery language that adds no substance.	"This red car combines **style, charm, and adventure** that captivates everyone."
Weasel Words	Employing vague qualifiers that dodge firm statements.	"Studies suggest our product **may** help improve results in **some** cases."
Paltering	Presenting literally true statements intended to mislead.	"**Historically**, the fund has demonstrated strong returns..." (omit the high risks)
Unverified Claim	Asserting information without evidence or credible support.	"Our drone delivery system enables **significant** reductions in delivery time."
Sycophancy	Offering insincere flattery and agreement to please.	"You're completely right; that's an excellent and insightful point."

definition, studying machine bullshit via two complementary perspectives: (1) introducing a quantitative *Bullshit Index (BI)* that quantifies indifference to truth; and (2) detecting prominent rhetorical phenomena previously identified in literature (Frankfurt, 1986; Bergstrom & West, 2021).

The Bullshit Index (BI). To operationalize Frankfurt's conceptual definition, we introduce the *Bullshit Index (BI)*, quantifying an AI model's indifference to truth. The BI measures how systematically a model's explicit claims depend on its internal beliefs. It is defined by two measurable signals: (i) the model's internal belief $p \in [0, 1]$ (probability placed on the statement being true), and (ii) its explicit claim $y \in \{0, 1\}$ (with 1 indicating "true"). Formally, the BI is the complement of the absolute value of the *point-biserial* correlation between the model's internal belief and explicit claims:

$$\boxed{\mathrm{BI} = 1 - |r_{pb}(p, y)|, \quad r_{pb}(p, y) = \frac{\mu_{p,y=1} - \mu_{p,y=0}}{\sigma_p} \sqrt{q(1-q)}}$$

where $\mu_{p,y=1}$ and $\mu_{p,y=0}$ represent the mean internal belief for statements explicitly claimed as "true" and "false" respectively, σ_p is the standard deviation of the internal belief distribution, and q is the proportion of explicit "true" claims. A BI close to 1 indicates that claims are largely independent of internal beliefs, revealing a high level of indifference toward truthfulness. Conversely, a BI near 0 implies that the model's claims strongly track its internal beliefs. To further distinguish sincerity from deliberate lying, we inspect the sign of $r_{pb}(p, y)$: a correlation near $+1$ indicates sincere alignment (truthful claims), while a correlation near -1 indicates systematic inversion (lying). Both extremes reflect low indifference, as claims remain closely tied to beliefs. Following prior work (Hendrycks et al., 2020; Jiang et al., 2020; Kadavath et al., 2022; Saunders et al., 2022; Azaria & Mitchell, 2023; Santurkar et al., 2023), we query the LLMs via multiple-choice question answering (MCQA) and measure the internal belief p as the probability assigned to the first token of the response.

Unlike prior work studying hallucinations focused on factual correctness, BI emphasizes *adherence* to the model's internal representation of reality. The BI thus differentiates honest-mistake hallucinations—adhering to inaccurate beliefs (low BI)—from bullshit hallucinations—disregarding internal beliefs (high BI). In Section 4, we compare BI scores before and after interventions (e.g., RLHF), providing empirical evidence aligned with Frankfurt's notion of bullshit as indifference to truth.

A Taxonomy of Machine Bullshit. In addition to quantifying indifference to truth, we build on the taxonomy of bullshit introduced by Bergstrom & West (2021) and operationalize their definitions so that we can measure them in the input–output behavior of LLMs.

- **Empty Rhetoric:** Text that is linguistically fluent and superficially persuasive but lacks substantive content. It initially appears meaningful but offers no actionable or factual insight.
- **Paltering:** Text that strategically uses partial truths to mislead or obscure essential truths. Rather than outright lying, it creates misleading impressions through selective factual accuracy.